

Dashboard Storyboard

MedTrack_DV – Hospital Analytics Dashboard

Prepared For: Infosys Springboard Virtual Internship (Data Virtualisation)

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Module: Module 4 – Dashboard Planning & Prototyping

Project Overview

Purpose of the Dashboard

The MedTrack_DV dashboard suite provides hospital stakeholders with a consolidated view of patient care, admissions, billing, and resource utilization. This storyboard documents the planned dashboard layouts, KPIs, charts, filters, and navigation to guide Tableau dashboard development..

Why Hospital Analytics Is Important

Hospitals generate large volumes of data across departments, wards, and patient records every day. A well-designed analytics dashboard makes this information accessible at a glance and supports faster, more informed decision-making across the hospital.

Objectives & Target Users

Objectives

- Design dashboards covering hospital-wide, patient-level, department-level, and resource-level analytics
- Plan KPIs, charts, and filters based on the available dataset before developing the dashboards in Tableau
- Ensure dashboards are intuitive for non-technical hospital staff
- Maintain a consistent, healthcare-appropriate visual design across all four dashboards

Target Users

Hospital Administrator	High-level view of hospital performance, occupancy, and patient volumes
Department Head	Department-specific patient and severity breakdowns for staff and ward planning
Management Team	Billing, payment status, and resource utilization views for financial monitoring

Dashboard Planning

Based on the hospital dataset— including patient demographics, admissions, billing, departments, severity, test results, wards, occupancy, payment status, and hospital regions— four dashboards were planned. Each dashboard includes interactive visualizations, KPIs, and filters.

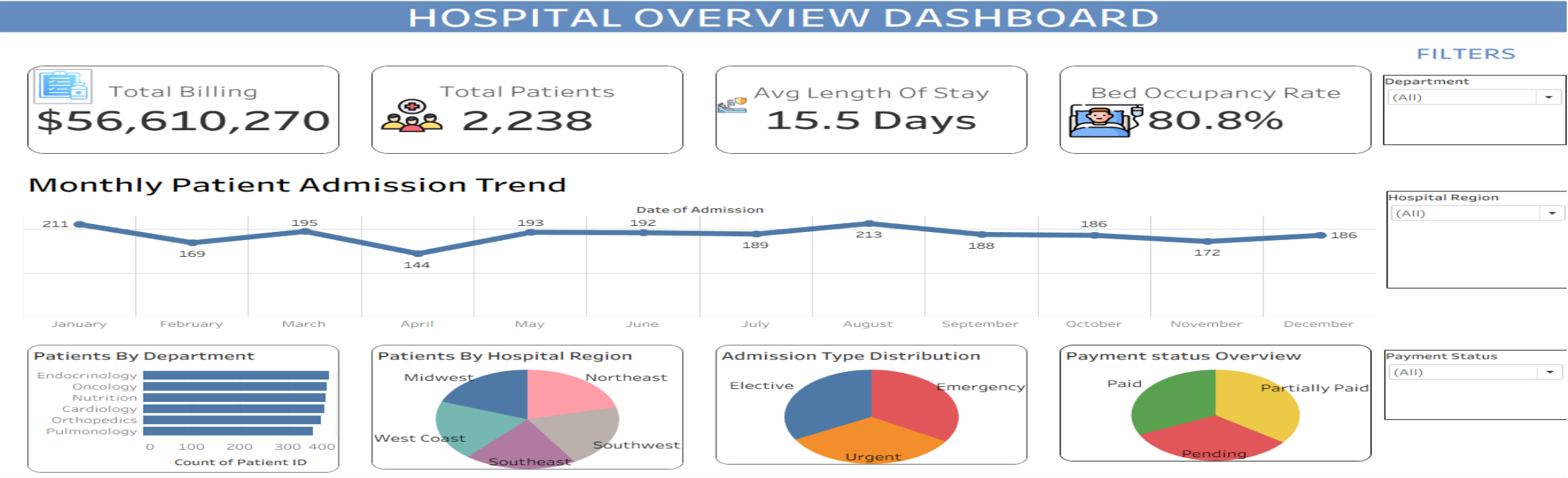
#	Dashboard	Primary Focus
1	Hospital Overview	Hospital-wide summary of admissions, occupancy, and regional distribution
2	Patient Flow	Patient journey from admission to discharge
3	Department Analytics	Department-wise patient load, severity mix, and billing patterns
4	Resource Utilization	Ward occupancy, bed utilization, and room distribution across the hospital.

Dashboard 1 – Hospital Overview

Purpose: Provides a high-level overview of hospital performance, including patient volume, occupancy, admissions, and regional distribution.

KPIs Planned	Charts Used	Filters Planned
Total Billing Total Patients Average Length of Stay Bed Occupancy Rate	Line Chart (Monthly Patient Admission Trend) Horizontal Bar Chart (Patients by Department) Pie Chart (Patients by Hospital Region) Pie Chart (Admission Type Distribution) Pie Chart (Payment Status Overview)	Department Hospital Region Payment Status

Navigation: Main dashboard providing an overview of hospital operations.

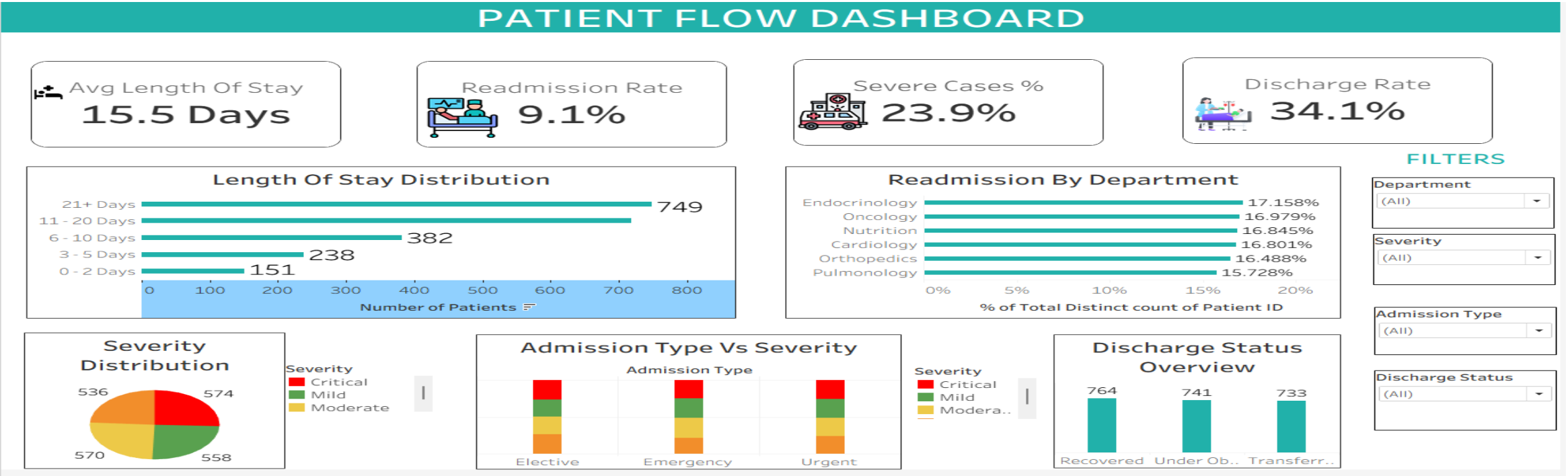


Dashboard 2 – Patient Flow

Purpose: Visualizes patient movement from admission to discharge, along with severity, readmission, and discharge patterns.

KPIs Planned	Charts Used	Filters Planned
Average Length of Stay Readmission Rate Severe Cases % Discharge Rate	Horizontal Bar Chart – Length of Stay Distribution Horizontal Bar Chart – Readmission by Department Pie Chart – Severity Distribution Stacked Bar Chart – Admission Type vs Severity Bar Chart – Discharge Status Overview	Department Severity Admission Type Discharge Status

Navigation: Focuses on patient movement and clinical flow across the hospital.

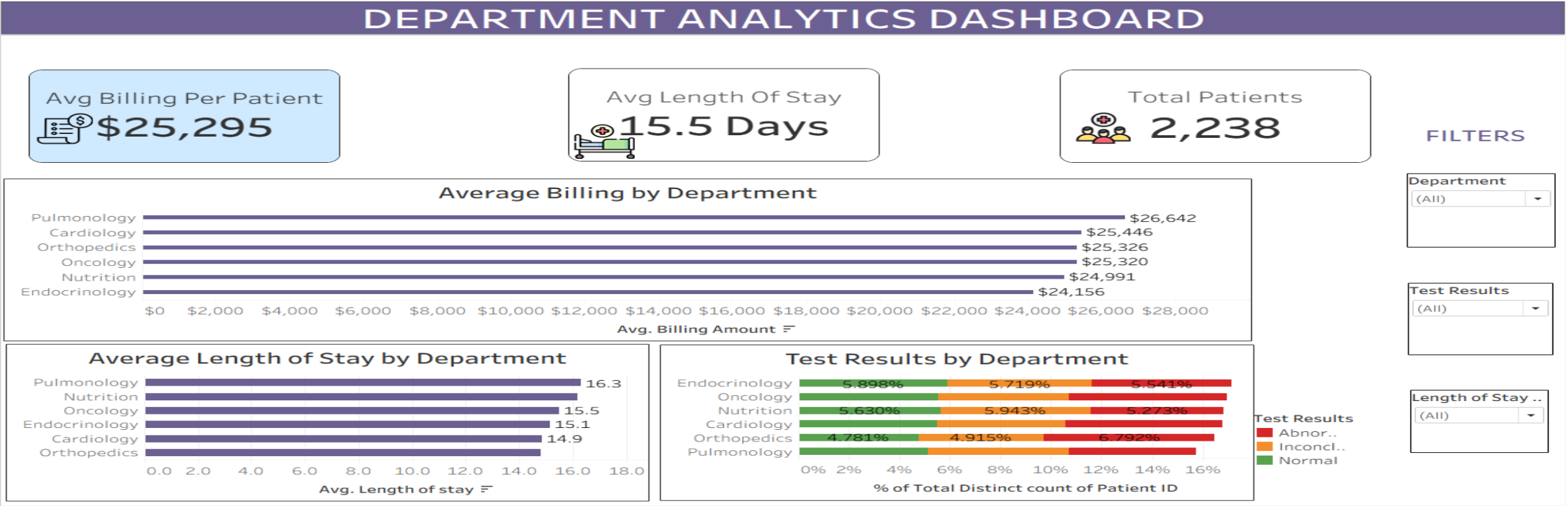


Dashboard 3 – Department Analytics

Purpose: Gives department heads a focused view of patient load, severity mix, and billing patterns within their department.

KPIs Planned	Charts Used	Filters Planned
Average Billing Per Patient Average Length of Stay Total Patients	Horizontal Bar Chart: Average Billing by Department Horizontal Bar Chart: Average Length of Stay by Department Stacked Horizontal Bar Chart: Test Results by Department	Department Test Results Length of Stay

Navigation: Provides department-level performance analysis.

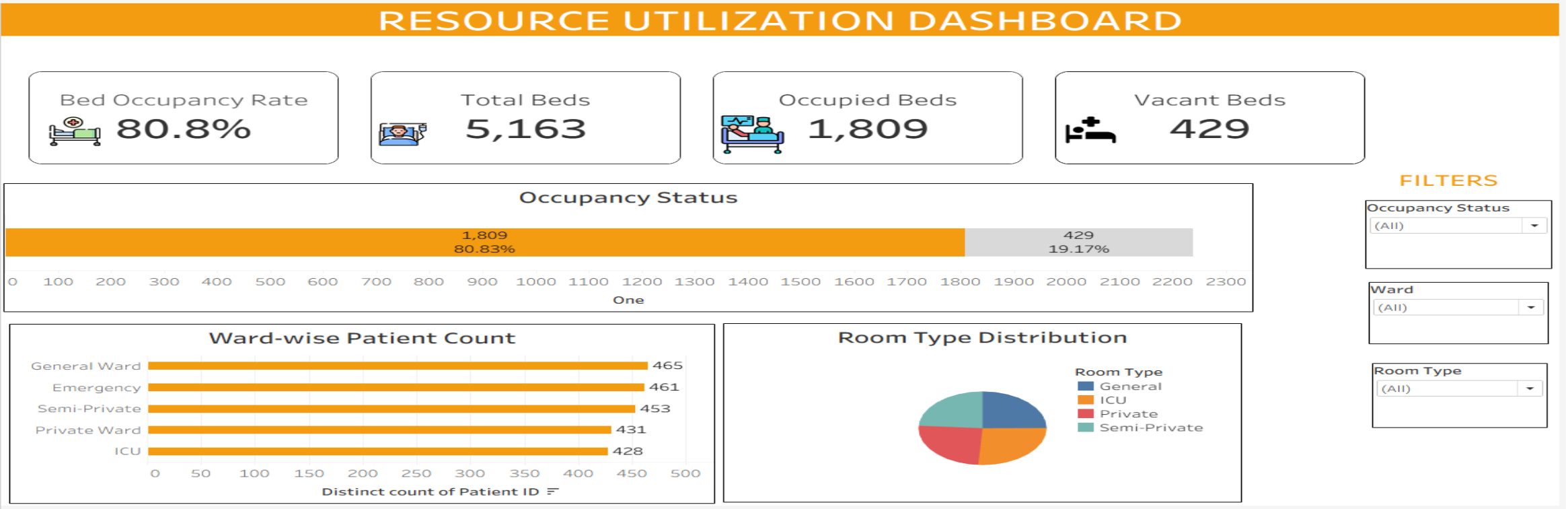


Dashboard 4 – Resource Utilization

Purpose: Tracks bed occupancy, ward utilization, and room distribution to monitor hospital resource utilization and operational efficiency.

KPIs Planned	Charts Used	Filters Planned
Bed Occupancy Rate Total Beds Occupied Beds Vacant Beds	Horizontal Bar Chart – Occupancy Status Horizontal Bar Chart – Ward-wise Patient Count Pie Chart – Room Type Distribution	Occupancy Status Ward Room Type

Navigation: Displays hospital resource and occupancy utilization.



Dashboard Interactivity

Planned prototype features – not yet implemented

Filters & Sorting

Global date range filter across all dashboards

Department, ward, and severity filters scoped to relevant dashboards

Payment Status filters for Hospital Overview and Occupancy/Ward filters for Resource Utilization.

Sortable tables (department/ward summary) by column value

Ranked bar charts sorted by metric value

Interactive Charts

Click-to-filter interactions between summary and detail views

Tooltips on hover showing underlying metric values

Navigation

Navigation buttons planned between the four dashboards

Planned drill-down from Hospital Overview to Department Analytics for department-level analysis.

Design Intent

All interactivity is planned with non-technical hospital staff in mind – clear labels, minimal clicks, and consistent filter placement across dashboards.

Design Considerations

Consistency	Same color palette, fonts, and KPI card style used across all four dashboards
Simple Layout	Clear top-to-bottom reading order: KPI cards first, then supporting charts
Healthcare-Friendly Visualization	Calm blue/white palette; bright colors reserved for flagging severity
Ease of Navigation	Consistent filter placement and navigation buttons across dashboards
Readable Charts	Clear titles, labeled axes, limited categories per chart to avoid clutter
Professional Structure	Each dashboard follows the same template: title, KPI row, chart grid, filter panel

Future Enhancements

Beyond the scope of the current planning module

Advanced KPIs – derived metrics such as cost per patient or department efficiency scores

Predictive Analytics – forecasting future admissions or occupancy using historical trends

Real-Time Monitoring – live data refresh for occupancy and admissions

Role-Based Dashboards – tailored views for administrators, department heads, and management

Drill-Down Analysis – deeper drill-down paths from hospital-level to patient-level detail

Alerts – automated notifications when metrics cross planned thresholds

Mobile Dashboard Support – responsive layouts for tablets and mobile devices.

Conclusion

This storyboard documents the planning and prototype design for the four MedTrack_DV dashboards – Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization. It Captures the intended dashboard layout, KPIs, visualizations, filters, and navigation flow for each dashboard, along with placeholders for screenshots once the dashboards are built.

This document forms the foundation for the full Tableau implementation to be carried out in the subsequent modules of the Infosys Springboard Data Visualization Virtual Internship.